

Stage 3: Model Building

C22 Retail Fade Strategy

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*Route taken: **report the negative result**. All figures are reverseProfit per lot; the \$7.00/lot cost is already inside the target and is applied exactly once (asserted in code, max deviation 9.09×10^{-13}). Hurdle is exactly zero. Common split: fit and select on C33–C52 ($n = 694$), evaluate on C53–C66 ($n = 6,583$). Our own split is reported alongside. All intervals are 95% bootstrap clustered by account (traderKey), with ipClusterId as a secondary check.*

1. Result

There is no exploitable edge at this cost level that we can demonstrate, and we can say how large an edge we would have detected if one existed.

- No admissible feature survives Benjamini–Hochberg correction across a 101-trial log, on either track or jointly. The only full-log survivors are tests of `sl_distance_pct`, which C22 has since ruled inadmissible.
- **Zero of seven** coverage levels, from 1% to 85%, produce an observed effect exceeding their own minimum detectable effect.
- The mandated training split cannot calibrate an executable threshold: maximum training support at any executable threshold is $n = 4$, against our pre-registered floor of $n \geq 30$.
- The frozen model shows +10.43/lot equal-weighted but –33.85/lot size-weighted, losing \$3,556.81 in absolute dollars against DO NOTHING's \$0.

Sections 2–6 evidence each claim. Section 7 states what we freeze and what we forecast.

2. Model, and why

Ridge on reverseProfit per lot, target winsorised at the 1st/99th percentile computed on training folds only. Inner expanding-campaign CV selected $\alpha = 1000$; we froze $\alpha = 3162.28$, one half-decade more regularised, because Stage 2 documented sharp cross-fold instability and a composite of weak signals should be shrunk harder than CV alone suggests. A regularised composite is the correct response to many weak, individually insignificant features — which, after the stop-loss exclusion, is exactly what remains.

Feature set: V2 (24 admissible features, trader-history fields dropped). We built and audited two variants. V1 retained the four cross-campaign history features with an explicit cold-start indicator; V2 dropped them. V1 was rejected on evidence:

- Under 79.39% cold-start in the evaluation window, `is_cold_start` carries a ridge coefficient of 2.705 against 0.012–0.547 for the four history features. V1 is fitting the cold-start partition, not trader history.
- Full-evaluation intervals overlap: V1 $[-49.44, 43.16]$, V2 $[-58.43, 37.92]$.
- On the 20.57% of rows that actually carry trader-specific history — the only place history can help — V1 *loses* to V2 (–2.58 vs +10.03).

V2 is preferred on parsimony, with no history-specific benefit demonstrated.

Inadmissible features. `sl_distance_pct`, `sl_widening_delta`, `has_sl` and `has_tp` are excluded per C22's ruling that `slPrice` is a close-time field. We attempted the recovery C22 invited — reconstructing the opening-ticket stop from the raw fills — and **disproved it**: across 4,543 multi-fill primary-era positions, `slPrice` changes between first and last fill in **73.12%** of cases (`tpPrice`, 79.84%); the collapse takes the first *non-null* value, so where a stop was added mid-trade that value is itself mid-trade; direction-violation rates are 11.98% on the first fill against 16.51% on

the last, both far above the near-zero rate a genuine open-time field would show; and violating rows are systematically profitable (−68.53/lot, 57.12% win rate), meaning the violation flag alone encodes outcome. The opening stop is not recoverable from this data. `sl_usage_rate_5`, `manual_exit_rate_5` and `trader_prior_sl_discipline` are retained per C22’s ruling; we note that the last measures a trader’s mean *mid-trade* stop state across prior campaigns, not stop-setting discipline at entry.

Decision rule, as executed by `predict()`:

$$\text{FADE} = (\text{score} > \max(\theta_{\text{sel}}, \text{hurdle})) \wedge (\text{win_streak} \leq 1) \wedge (\text{trades_per_hour} \leq 60)$$

with $\theta_{\text{sel}} = -10$ and `hurdle` = 0. The two abstain overrides are carried forward from Stage 1–2 findings and were fixed before this stage.

3. Backtest on the common split

	<i>n</i> acted	coverage	equal-wtd	size-wtd	total \$
Model (overrides)	177	2.69%	+10.433	− 33.849	−3,556.81
Model (no overrides)	183	2.78%	+10.452	—	−3,534.65
DO NOTHING	0	0.00%	0.000	0.000	0.00
FADE EVERYTHING	6,582	100.00%	+5.436	−3.001	−12,108.71

Model account CI [−52.786, 72.331], IP CI [−45.183, 66.523]; size-weighted account CI [−74.360, 1.166]. FADE EVERYTHING equal-weighted CI [−9.339, 20.713], size-weighted CI [−11.031, 5.381]. Dollar totals exclude C66 (see data note), so C53–C65 and C53–C66 figures are identical here; the power analysis in Section 5 retains C66’s per-lot observation, giving $n = 178$ and +10.325.

The two metrics disagree, and the dollar outcome is the honest one. The model captures **29.37%** of the total fade-everything dollar loss while acting on 2.69% of positions and 2.60% of total volume — $10.92\times$ its position-coverage share and $11.28\times$ its amount share. It is selecting large-lot losers. Against DO NOTHING’s \$0 it loses \$3,556.81. We report the equal-weighted figure because it is the requested metric, and the size-weighted figure because it is what happens to the book.

Per-campaign. The result is not distributed. C53 gives +173.557 on $n = 5$ (the only positive campaign whose CI excludes zero, [48.512, 510.000]); removing it halves the result to +5.691 ($n = 172$, CI [−60.200, 66.313]). C59 is significantly *negative*: −99.990 on $n = 21$, CI [−221.813, −8.205]. Seven of thirteen campaigns are negative.

Our own split (expanding-window folds within C53–C65; Track A trader-purged, Track B unpurged) reproduces the pattern and adds one failure the common split hides. On Track A the model returns +14.130 ($n = 114$, 5.21%, CI [−80.926, 111.031]) against fade-everything’s +21.711 (CI [−3.458, 44.978]) — **a lower point estimate and a wider interval than the baseline it must beat**. Track B fold 2 is significantly negative (−210.475, CI [−479.776, −43.912]), and Track B fold coverage swings 5.58%, 1.84%, 16.14%, 0.50%. The threshold is not stable across campaign blocks.

In-sample to out-of-sample gap. Training +50.897 on $n = 4$ (CI [−62.924, 193.331]); evaluation +10.325 on $n = 178$. The gap is −40.571/lot, but the training figure rests on four observations and is not a meaningful anchor — which is itself the subject of Section 4.

4. The mandated split cannot calibrate a threshold

This is a structural property of the common split, not a tuning failure, and we report it because it bounds what any submission can conclude from this split.

- **The training window is 694 positions across 330 accounts** against 6,583 evaluation positions — a train set roughly one-tenth the size of the test set. That is 28.92 positions, or 13.75 accounts, per admissible feature.

- **79.39% of evaluation positions belong to traders with no history in the training window.** Four history features are structurally uncomputable for four in five evaluation rows. This is why V2 drops them.
- **Training MDE is \$65–78/lot** (ICC 0.162 / DEFF 1.753 by account; 0.096 / 1.615 by IP). Only **1 of 48** admissible pooled effects observed anywhere in Stage 2 exceeds that floor, so the mandated training split could not have detected 47 of the 48 effects we observed — a null result on it is uninformative rather than evidence of absence. Fitting across the C53 regime break costs a further \$12.27/lot versus training within the post-break era, measured directly.
- **No executable threshold has adequate support.** Selection maximised training mean subject to $n \geq 30$ and chose $\theta = -10$ ($n = 88$, 12.68% of the training window). But the economic hurdle forbids fading a negative predicted return, so every negative threshold is overridden and the true operating point is $\text{score} > 0$, where training support is $n = 4$. Re-running selection over the executable grid $[0, 500]$ confirms the maximum available support at any executable threshold is $n = 4$, with training mean $+50.897$ — the same four observations reported as the in-sample result in Section 3. **The pre-registered $n \geq 30$ floor is unsatisfiable on this split.** The threshold sweep is therefore inert by construction: execution reduces to $\text{score} > 0$ plus the two abstain overrides.

5. Power: what we would have detected

Target	n acted	achieved	observed rP/lot	account CI	MDE (floored)	clears?
1%	66	1.00%	+0.918	$[-116.23, 110.63]$	191.50	no
2.7%	178	2.70%	+10.325	$[-52.20, 72.29]$	116.61	no
5%	329	5.00%	+23.329	$[-27.05, 75.82]$	92.28	no
10%	658	10.00%	+21.691	$[-21.42, 66.44]$	71.30	no
25%	1,646	25.00%	+31.743	$[-0.61, 62.69]$	43.41	no
50%	3,292	50.01%	+21.666	$[-0.20, 43.67]$	31.53	no
100%	5,616	85.31%	+13.135	$[-2.93, 29.41]$	26.49	no

Coverage levels are produced by sweeping the execution hurdle while retaining both abstain overrides; the frozen model sits at the 2.70% row, and coverage caps at 85.31% because the overrides are unconditional. $\text{MDE} = 2.801585 \sigma \sqrt{\text{DEFF}/n}$ at $\alpha = 0.05$, power 0.80, with σ held at the primary-era dispersion (raw 555.33; winsorised 482.45) so the calculation does not reuse the selected subset's realised variance. Column shows account-clustered raw- σ MDE with ICC floored at zero. Acting-subset ICC estimates are negative (-0.089 account, -0.098 IP) at mean cluster size ≈ 2.7 ; the resulting $\text{DEFF} < 1$ is a small-sample artefact and anti-conservative, and the training-window ICC is positive (0.162 account), so the two are not comparable without the floor. Unfloored account MDE at 2.70% is \$107.46 raw / \$93.36 winsorised. Zero of seven rows clear either the floored or the unfloored MDE.

Zero of seven coverage levels clear their own detection floor. The closest approach is 25% coverage, at a ratio of 0.84 observed-to-MDE on winsorised σ . This is the general result: narrowing coverage to find a state where edge holds raises the detection floor as $1/\sqrt{n}$, faster than narrowing plausibly raises the effect. At this cost level and sample size there is no coverage band where power and edge are simultaneously satisfied.

The score does not rank trades. If it ordered by fadeability, mean edge would decline monotonically as lower-scored trades are added. Instead the top 1% of scores returns $+0.918/\text{lot}$ — below every broader operating point, and on the widest interval in the table — while mean edge peaks at 25% coverage rather than falling. The evaluation score distribution (median -29.92 , p75 -15.29 , p99 $+27.53$, s.d. 30.36) shows the model predicting a loss on three-quarters of the window: an independent statement, from a fitted model rather than a raw average, that fading is unprofitable here.

6. Trials log and multiple-testing correction

101 trials logged across Stages 1–3; 90 scalar outcome tests corrected. Under Benjamini–Hochberg at $q = 0.05$: **no admissible Stage 2 feature survives**, by track or jointly. Across the full log, the

only survivors are tests of the now-inadmissible `sl_distance_pct` — **the leak was producing our only statistically defensible results.**

Six conditional states were pre-registered before running and tested under two threshold provenances: S1, thresholds selected on training campaigns ≤ 52 only (underpowered, MDE \$65–78 at $n = 694$); and S2, thresholds carried forward from Stage 1–2 findings, which were derived on C53–C65 and are therefore leaky under the mandated split. No state survives correction under either. S2 states 3 and 4 clear zero unadjusted — we report them as the negative control that demonstrates why the trials log exists, not as a finding.

7. What we freeze, and what we forecast

Frozen artefact: `artifacts/stage3_v2.json` (stage3-v2-2026-08-20), V2 features, $\alpha = 3162.28$, hurdle 0, both abstain overrides. Entry point `predict(trades_df)`; deterministic, pinned dependencies, one documented command, with a banned-column leakage assertion and a truncated-sequence recompute check in the test suite (19 tests). We freeze it **as-is and disclosed as uncalibrated**: the operating point has $n = 4$ training support and cannot meet our own pre-registered floor. An always-abstain model would tie DO NOTHING rather than beat it, and would generate nothing for Stage 4 to score.

Disclosed post-hoc diagnostic, not frozen. The size-weighted loss is concentrated in large lots, so a size cap in the decision rule would improve the dollar outcome. We did not adopt it: that rule was chosen after observing the size-weighted result, and adopting it is precisely the behaviour the trials log exists to catch. It is reported as a variant and logged as a trial.

Stage 4 forecast. Registered after the model was frozen and evaluation results were viewed.

Equal-weighted mean reverseProfit per lot on unseen campaigns: **0.00**, 95% interval $[-62, +62]$. Coverage: **under 5%**, threshold-driven and unstable across campaign blocks (0.50%–17.89% across our own folds). Size-weighted mean: **negative**. We expect the model **not** to beat DO NOTHING in absolute dollars.

Derivation. The point estimate is zero because nothing in the analysis supports any other value: no feature survives correction, and no coverage level produces an effect above its own MDE. The interval is the half-width of the acting-subset account-clustered CI $[-52.20, +72.29]$, half-width 62.25, re-centred on zero; removing the C53 outlier does not tighten it (63.26). We forecast a null rather than the observed +10.33 precisely because our own analysis shows +10.33 is indistinguishable from zero.

8. Where it fails

1. **Size selection.** Equal-weighted positive, size-weighted -33.849 . The model concentrates into large-lot losers — the Stage 1 sizing paradox reappearing at the decision layer.
2. **Calibration.** The operating point rests on four training observations. Any threshold shift is unsupported by data.
3. **Concentration.** Half the equal-weighted result comes from one five-trade campaign.
4. **Instability.** Track B fold coverage ranges 0.50%–17.89%; fold 2 is significantly negative.
5. **No ranking power.** The top 1% of scores returns +0.918/lot, below every broader operating point; mean edge peaks at 25% coverage rather than declining monotonically.
6. **Regime dependence.** Training across the C53 break costs \$12.27/lot, and Stage 4 campaigns lie further still from the training window.

Data note. C66 collapses 1,911 fills into a single position whose *amount* is the sum across fills (462.30 lots, against a maximum collapsed amount of 6.50 lots in C53–C65), contributing $-\$4,019.40$. Sum aggregation remains correct for ordinary partial-close fills and no other primary-era multi-fill position is affected; C66 is a corrupted test export and is excluded from every dollar total, while its per-lot observation is retained in the $n = 178$ power comparison. Cost application was asserted exactly once in code, so no Stage 2 restatement is required.